1. Business context

Harbour & Co. is an Australian-owned small-to-medium e-commerce retailer specialising in lifestyle and home-office products. The company was established in 2022 and has grown primarily through online sales. It currently sells approximately 150–200 products across several categories, including Home Office, Desk Accessories, Lighting, Storage & Organisation, Technology Accessories, and Lifestyle Products. Harbour & Co. serves customers across Australia, with sales generated through its own website as well as third-party marketplaces such as eBay and Amazon Australia. Its customer base consists of university students, young professionals, home-office workers, small businesses, and individual consumers.

The company has experienced relatively strong growth over the past two years. The number of orders, customers, and products sold has increased, while the business has expanded its presence across multiple online sales channels. From a high-level perspective, management considers the business to be performing well because overall revenue and order volumes have continued to increase. However, as the business has become larger and more complex, management has become increasingly concerned that revenue growth alone may not provide an accurate picture of the company's overall performance. In particular, management has noticed that some periods with strong sales do not appear to generate a proportionate increase in profitability.

One of the main challenges is understanding the difference between **sales performance and actual profitability**. A product may generate a large amount of revenue because it has a high sales volume, but its profitability may be relatively low after considering product costs, discounts, marketplace fees, shipping costs, and other expenses. Conversely, some products may have relatively low sales volumes but generate higher profit margins. Management currently relies heavily on revenue and order volume when evaluating product performance, and they are unsure whether this approach is identifying the products that create the greatest value for the business.

The company's use of discounts and promotional campaigns has also created uncertainty. Harbour & Co. regularly uses discounts ranging from approximately 5% to 20% through seasonal promotions, marketplace campaigns, and special offers. These promotions are intended to increase customer demand, attract new customers, compete with other online retailers, and reduce slow-moving inventory. While some campaigns have successfully increased order volumes, management does not have a clear understanding of whether higher sales generated through discounts are translating into meaningful improvements in profitability. For example, a promotion may increase the number of orders by a significant percentage while producing only a small increase in profit. Management therefore wants to understand whether its current discounting strategy is creating sustainable value or simply increasing sales volume at the expense of margins.

Harbour & Co. is also experiencing differences in performance across its sales channels. The company's website gives it greater control over pricing, promotions, customer

relationships, and customer data, while marketplaces such as eBay and Amazon provide access to a larger customer base. However, marketplace sales are also associated with additional fees and competitive pricing pressures. Management has observed that some products perform particularly well on marketplaces in terms of sales volume, but they are uncertain whether these products generate the same level of profitability as products sold through the company's own website. At present, management does not have a consolidated analytical view that allows them to easily compare revenue, profitability, discounts, and customer performance across different sales channels.

The company's product portfolio has also become increasingly difficult to manage. With approximately 150–200 active SKUs, some products sell very quickly while others remain in inventory for extended periods. High-demand products may occasionally experience low-stock situations, potentially resulting in missed sales opportunities, while slow-moving products tie up working capital and warehouse space. Management does not want to increase inventory across all products because doing so would increase storage and working-capital requirements. Instead, they would like to understand which products appear to have strong demand, which products may be underperforming, and whether sales and profitability patterns can provide useful information for future inventory decisions.

Customer behaviour is another area where management believes there may be untapped opportunities. Harbour & Co. has accumulated transaction data that allows it to identify new and returning customers, but this information is not currently being used systematically to understand customer value. Some customers make a single purchase, while others purchase repeatedly throughout the year. Management wants to better understand whether a relatively small group of customers contributes a disproportionate amount of revenue and whether returning customers demonstrate different purchasing behaviour from one-time customers. However, the company currently lacks a clear analytical framework for identifying its most valuable customer groups.

Geographic performance also varies across Australia. Harbour & Co. receives orders from Victoria, New South Wales, Queensland, Western Australia, South Australia, Tasmania, the Australian Capital Territory, and the Northern Territory. Management has observed differences in sales volumes between states, but revenue alone does not necessarily indicate which markets are most attractive. Differences in shipping costs, order sizes, discounts, and product preferences may affect the profitability of customers in different locations. As a result, management is interested in understanding whether some geographic markets are generating strong revenue but relatively weak profitability, and whether there are particular locations that may represent opportunities for future growth.

The company's current reporting process makes these issues more difficult to investigate. Data is collected from several parts of the business, including the company's website, eBay, Amazon, warehouse operations, finance, and marketing activities. Different teams often maintain their own Excel or CSV reports, which means that information about sales, products, customers, costs, inventory, and promotions is distributed across multiple sources. Monthly reporting is largely manual, and management does not currently have a single analytical view of overall business performance. As a result, managers can see individual numbers but often struggle to connect those numbers together and determine what is actually driving changes in business performance.

The data available for analysis consists of approximately two years of historical transaction records. The dataset contains information relating to orders, customers, products, sales channels, product costs, selling prices, quantities, discounts, shipping costs, customer locations, promotional campaigns, and return status. However, the data is not perfectly clean. It contains issues such as missing customer information, duplicate records, inconsistent product and category names, different formats for dates and locations, unusual quantities, and negative quantities associated with returned products. Some records may also contain missing information about promotional campaigns or costs. As a result, an important part of the analysis will involve assessing data quality and determining appropriate methods for cleaning and preparing the dataset before conducting the business analysis.

Harbour & Co. does not have a large internal data analytics team and is looking for a practical solution rather than a highly complex analytical model. Management does not expect the analysis to involve advanced machine learning or sophisticated predictive modelling. Instead, they want a clear and repeatable analytical process that can help them understand what is happening within the business, identify potential performance gaps and opportunities, and support better decision-making. Ideally, the final solution should be simple enough to update periodically as new transaction data becomes available.

You have therefore been engaged as a **Junior Business Analyst** to conduct an analytical review of Harbour & Co.'s historical business data. Management has provided access to the available transaction, product, customer, sales channel, and promotional data but has not specified exactly what analyses should be performed. Instead, they expect you to investigate the data, identify the most important business issues, and determine which findings could provide meaningful value to management. Your final analysis should move beyond simply describing sales figures and should help management understand **what is happening, why it may be happening, where the major opportunities or risks exist, and what actions the business could consider taking as a result.**

2. Business Problem

Harbour & Co. currently lacks a clear, efficient and consolidated way to understand and connect its business data across different departments and sales channels. The manual and fragmented reporting process makes it difficult for management to assess whether increasing sales and order volumes are translating into actual profitable growth. In particular, management has limited visibility into the factors that may influence profitability, such as product performance, discounting and promotional campaigns, sales channels, customer behaviour and geographic markets. As a result, the business may struggle to identify which products and channels generate the greatest value, determine where resources and investment should be prioritised, and distinguish genuine growth opportunities from increases in sales volume that may not generate proportionate profits. Without a more effective analytical process, management may make decisions based on incomplete or disconnected information, potentially limiting profitable growth and causing the business to overlook important performance gaps and opportunities.

3. Project Objective

The objective of this project is to analyse Harbour & Co.’s historical business data to identify the key factors influencing its sales performance and profitability. The analysis will examine

product, sales channel, customer, geographic and promotional performance to determine where the business is generating value, where performance gaps may exist, and which areas have potential for profitable growth. The project will translate these findings into practical, data-driven recommendations that can help management improve performance evaluation, prioritise resources and make more informed business decisions.

4. Key Analytical Questions

Is revenue growth translating into profitable growth?

Which products are driving profitable growth?

How do sales channels differ in profitability?

Is discounting supporting or reducing profitable growth?

Which customer/geographic segments provide attractive opportunities?